Two-mode Network Analysis of Culture Concepts

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Abstract

Kroeber and Kluckhohn's classic 1952 monograph *Culture: A Critical Review of Concepts and Definitions* brought together over a hundred scholarly definitions of culture and indexed the key concepts they discuss. We build two networks from that index to explore how concepts and works are connected. In the *definitions network*, concepts like "society" and "beliefs" are linked to 170 different scholars' definitions, grouped into seven categories but each credited to its own author. In the *works network*, the same concepts are linked to the books and papers cited in those definitions. By comparing both networks, we can see which concepts span many scholars' work and which are concentrated in particular sources. We present basic statistics about the networks, visualize the data using correspondence analysis and spanning trees, and highlight the key concepts and works that anchor each network.

1 Data and networks

We build two networks from K&K's Index B. The *definitions network* links 472 concepts—concrete ideas like "society," "customs," and "beliefs"—to 170 scholarly definitions of culture. Each definition carries the name of its author and falls into one of seven categories, which K&K label with the capital letters A through G: A Descriptive, B Historical, C Normative, D Psychological, E Structural, F Genetic, and G Incomplete. The *works network* links the same 472 concepts to 133 authored works (books and journal articles), so we can see which ideas appear in which scholarly sources. Both networks use the same set of concepts, which lets us compare how the definitions and works overlap in what they discuss. The definitions in the network come from K&K's own author index (167 sources) plus three additional attributions from the main text of their book (C-I-18a, D-II-16, E-17) but missing in the index. Concepts have consistent names across the networks; where a concept appears under different headings in K&K's source text, we treat it as a single concept. Works are named by their author(s) and year, and a single work can be cited under multiple definitions.

The definitions network naturally breaks into 11 separate components, with the largest containing 624 nodes. The works network has 10 components, with the largest containing 593 nodes. Our analysis focuses on these largest components, which contain virtually all the meaningful connections. We use maximum spanning trees across multiple projections to identify the strongest connections in different views of the network.

2 Overview of the networks

Table 1 gives summary statistics for both networks. The first three rows show node and edge counts. The "degree" rows counts connections of each node type; a concept's degree is how many

definitions (or works) mention it, while a definition's or work's degree is how many concepts it contains. In both networks, concepts and definitions differ substantially in their average degree.

Table 1: Summary statistics for the two networks. Average nearest-neighbor degree (k_{nn}) is the average degree of a node's neighbors; degree correlation shows how a node's degree relates to the degree of its neighbors.

	Definitions network	Works network
Concept nodes	472	472
Definitions / works	170 codes (5 isolated)	133 works (4 isolated)
Edges	1193	1165
Density	0.015	0.019
Mean degree, concepts	2.53 (range 1–36)	2.47 (range 0–33)
Mean degree, definitions/works	7.02 (range 0–25)	8.76 (range 0–51)
Mean k_{nn}, concepts	10.28	14.64
Mean k_{nn}, definitions/works	8.96	8.73
Degree correlation r	−0.096	−0.183

We see a notable pattern in the lower rows of the table documenting the degree correlation in each network; concepts with many connections tend to link to definitions or works with few, and vice versa. The nearest-neighbor rows show the other side of this pattern. A concept's neighbors average 10.3 (definitions network) or 14.6 (works network) uses, while a definition's or work's neighbors average only about 9 concepts, so concepts attach preferentially to heavily used nodes. The works network is also slightly denser (0.019 against 0.015) despite nearly identical edge counts, because it has fewer second-mode nodes. Finally, the maximum work degree of 51 belongs to Kelly & Kluckhohn 1945a, a fact that returns in the spanning-tree analysis below.

Figure 1 shows the degree distribution plots for each set of nodes in each network. Most concepts are lightly connected, with half appearing in only one or two definitions, and half being cited by only one or two works. In contrast, definitions and works are more heavily connected, wtih the average definition covering 7 concepts (median 7), and the average work covers 8 concepts (median 8), with some definitions and works discussing many more. This asymmetry means that a few concepts are discussed widely, while most concepts appear in just one or two places. The two second-mode distributions differ in their tails: works reach degree 51 while definitions top out at 25, so a single work can cover far more concepts than any single definition. The concept distributions, by contrast, are nearly identical across the two networks, as expected given that both share the same 472 concepts.

3 Maximum spanning trees

A maximum spanning tree (MST) keeps the strongest ties in a network while still connecting every node, using as few links as possible. A two-mode network has no ties between two concepts or between two definitions, so we analyze it through multiple one-mode projections [Everett and Borgatti, 2013]. Following Breiger [1974], two concepts are tied if they share at least one definition or work, with weight equal to the number they share. Likewise, two definitions (or works) are tied through the concepts they share. In matrix form, the projections are:

$$W_{\text{concept}} = AA^{\top}, \qquad W_{\text{def}} = A^{\top}A,$$

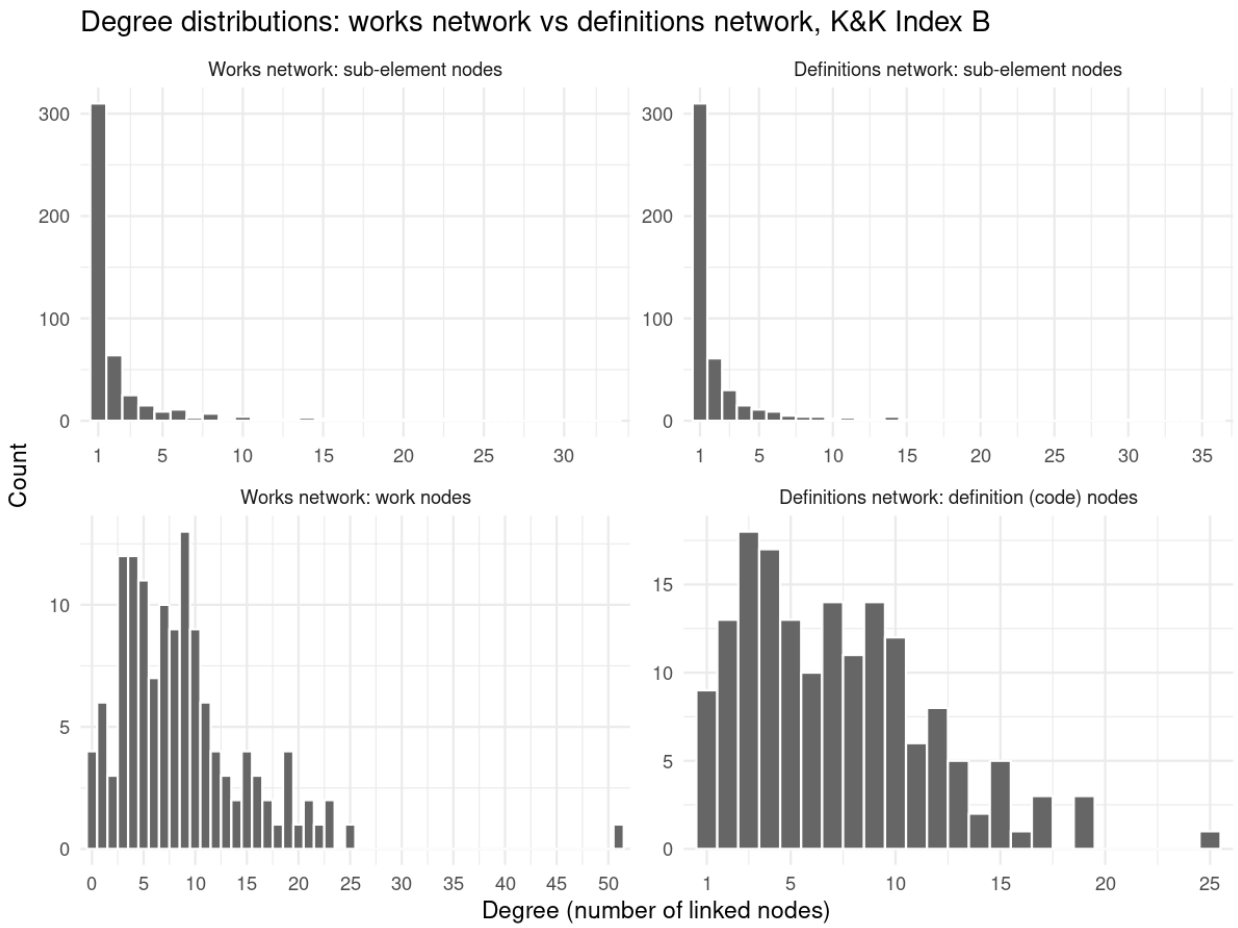


Figure 1: Degree distributions. Top row: sub-elements in the works network (left) and the definitions network (right). Bottom row: works (left) and definition codes (right).

where $(W_{\text{concept}})_{ij} = \sum_k a_{ik}a_{jk}$ counts the definitions that include both concepts i and j. Each projection is a weighted graph whose edges are the off-diagonal pairs with positive weight. We build projections from the 2-core of each network's largest component.

The maximum spanning tree of each projection is found by computing the minimum spanning tree on negated weights, $-w$. Kruskal [1956] sorts edges from largest weight to smallest and adds each unless it would close a cycle, producing a tree that maximizes $\sum_{e \in T} w_e$. Because the 2-core is connected, each tree has exactly $|V| - 1$ edges. In the visualizations, node size shows betweenness within the tree and node color shows the community structure detected using Newman's leading-eigenvector method applied to the tree.

The definitions network has two main projections. The concept-level tree (Figure 2) shows how 159 concepts cluster. The definition-level tree (Figure 3) shows how 145 definitions cluster. Table 2 lists the top 10 concepts and definitions by betweenness in their maximum spanning trees.

In the concept tree (Figure 2), the four largest nodes—social, group, society, and behavior—are exactly the top four of Table 2, and each anchors one of the tree's visible color regions. Social heads a green region of social-organizational vocabulary (social heritage, institutions, customs, social group), society anchors a pink region of learned and shared behavior (learned behavior, habits, socially transmitted), and behavior and group bridge the two. Beliefs and customs, fifth

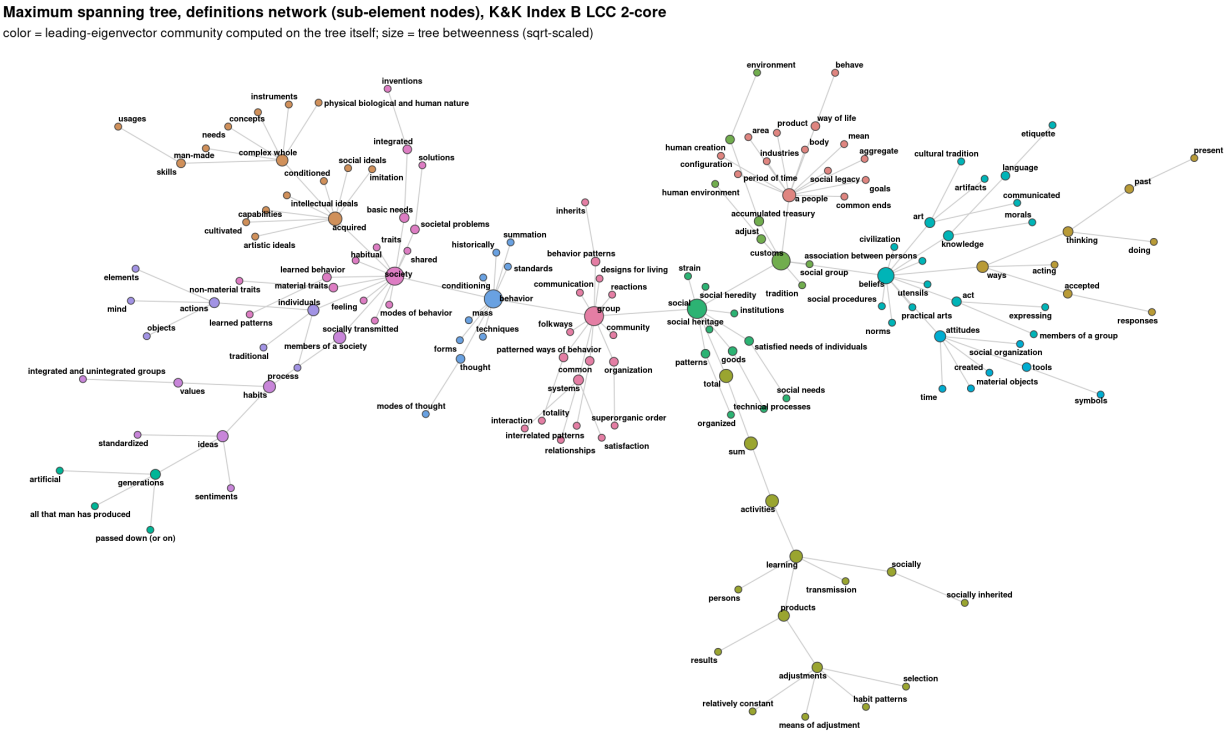


Figure 2: Maximum spanning tree of the definitions network, sub-element (concept) nodes. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

and sixth in the table, head the teal region of norms, knowledge, and artifacts. The tree also has one long peripheral branch running from total and sum down through activities, learning, transmission, products, and adjustments; this chain collects the acquisition vocabulary of the Genetic definitions and hangs off the core through a single path, so these ideas connect to the rest of the network only transitively.

The definition tree (Figure 3) puts its three largest nodes, F-I-3, C-I-6, and F-I-8, at the center of a single large community that mixes Normative (squares) and Genetic (pentagons) definitions, consistent with the table, where seven of the top ten definitions carry a C or F code. Degree and betweenness dissociate in this ranking: C-I-6 combines the second-highest betweenness with the highest tree degree (12), acting as the main hub of that community, while F-I-3 reaches the highest betweenness with only four ties, sitting on the paths that join the community to the rest of the tree. One author supplies a striking share of the bridges, as Bidney holds three of the ten most central definitions (C-II-3, C-II-4, and A-16). Away from the center, the Descriptive (A) definitions form a separate orange community in the upper left around A-3, and a chain of Descriptive and Historical definitions (A-16, C-I-4, A-1, A-14) runs along the right side, showing that the earliest and simplest definitions connect to the core through fewer, longer paths.

The works network also has two main projections. The concept-level tree (Figure 4) shows how 157 concepts cluster within the works network. The work-level tree (Figure 5) shows how 119 works connect through shared concepts. Table 3 lists the top 10 concepts and works by betweenness centrality.

The works-network concept tree (Figure 4) resembles its definitions-network counterpart but with a sharper hierarchy. Social is again the largest node, and here it sits at the junction of three large color regions, a green region of normative and symbolic vocabulary headed by customs

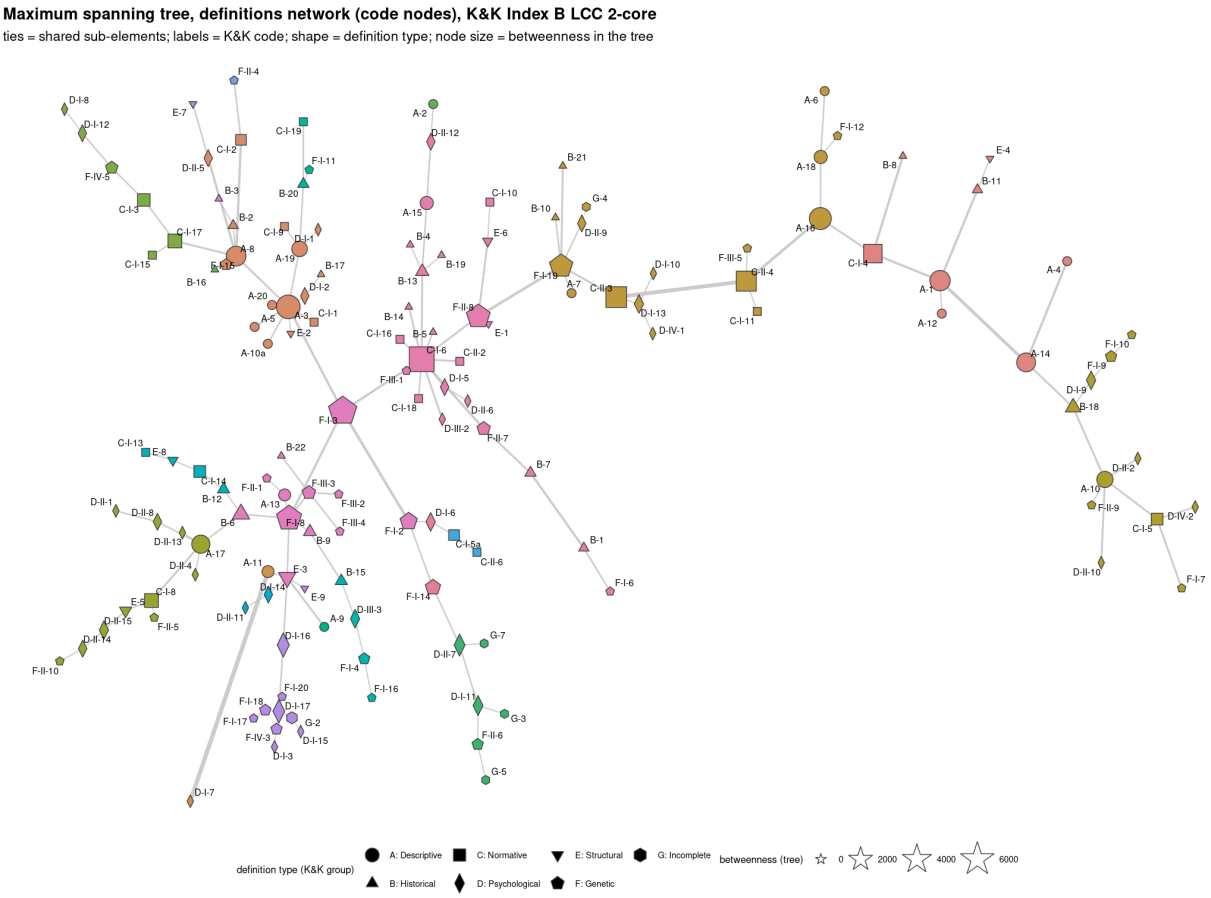


Figure 3: Maximum spanning tree of the definitions network, code (definition) nodes. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

(knowledge, beliefs, language, attitudes, symbols), a pink region of social organization around group, and a yellow region of behavior and acquisition (behavior, area, mass, transmission, habits). Society combines its high betweenness with the largest tree degree in the table (22), drawing many short spokes, while the acquisition chain seen in the definitions tree reappears at the top of the plot through members of a society, transmission, habits, and products.

The work tree (Figure 5) has a single dominant center. Kelly & Kluckhohn 1945a, the largest node and top entry of Table 3, sits at the middle of the plot with nineteen spokes reaching into every color region, so nearly any path between two works on opposite sides of the tree passes through it; it is at once the largest hub and the strongest bridge. Young 1934, Gillin 1942, Winston 1933, and Kroeber 1948a, the next four works in the table, appear as secondary hubs each heading their own community, and Linton appears twice in the top ten (1936 and 1945b), the only author with two such works. Most works, by contrast, are small peripheral leaves attached by a single edge, visible as the many small nodes at the plot's edges; Willey 1929, for example, despite its central position in the drawing, is one such leaf, a reminder that the spring layout places nodes by topology rather than importance.

Note to Table 2: Definition codes use the format X-Y-Z, where X is a letter (A–G) for K&K's seven definition categories (Descriptive, Historical, Normative, Psychological, Structural, Genetic, Incomplete), Y is a roman numeral for the subcategory, and Z is the number of the definition within it.

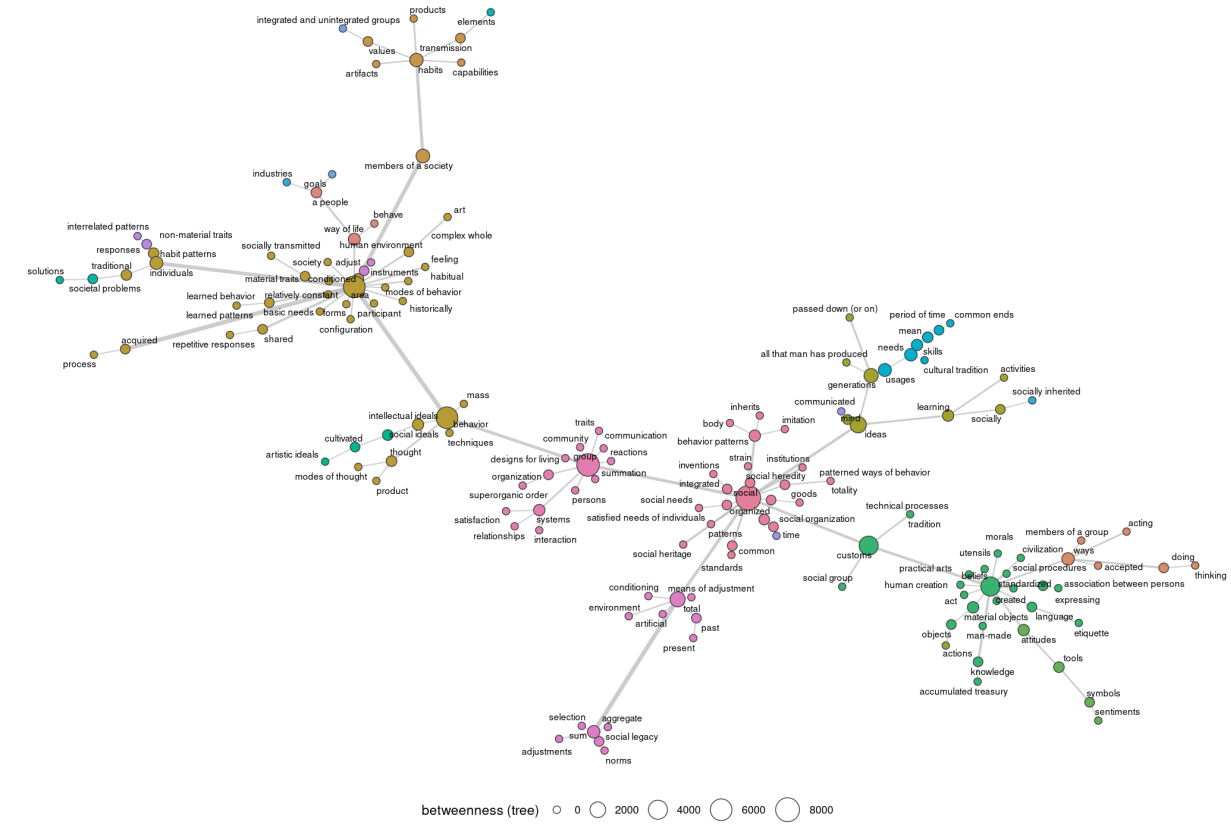


Figure 4: Maximum spanning tree of the works network, sub-element (concept) nodes. Ties count the works that cite both sub-elements.

Taken together, the four trees tell a consistent story. Eight of the ten most central concepts are the same in both networks—social, group, society, behavior, beliefs, customs, total, and members of a society—and each of the leading concepts anchors one of the color regions of the concept trees, so the conceptual core of the culture concept is stable whether it is measured on definitions or on the works that cite them (systems and sum enter the definitions ranking, ideas and generations the works ranking). The two networks differ in how concentrated their second mode is. Among definitions, centrality is spread fairly evenly: the top definition and the top concept reach almost identical normalized betweenness (69%), and no single definition approaches gatekeeper status. Among works, centrality concentrates at a single point, Kelly & Kluckhohn 1945a at 83% of the maximum possible, with the runner-up below 42%; the works literature is more star-shaped around one synthetic review than the definitions are around any of their members. In both concept trees the acquisition and transmission vocabulary sits on a single peripheral chain rather than at the core, suggesting that Genetic concerns, however common in the definitions themselves, connect to the rest of the conceptual structure only through a narrow path.

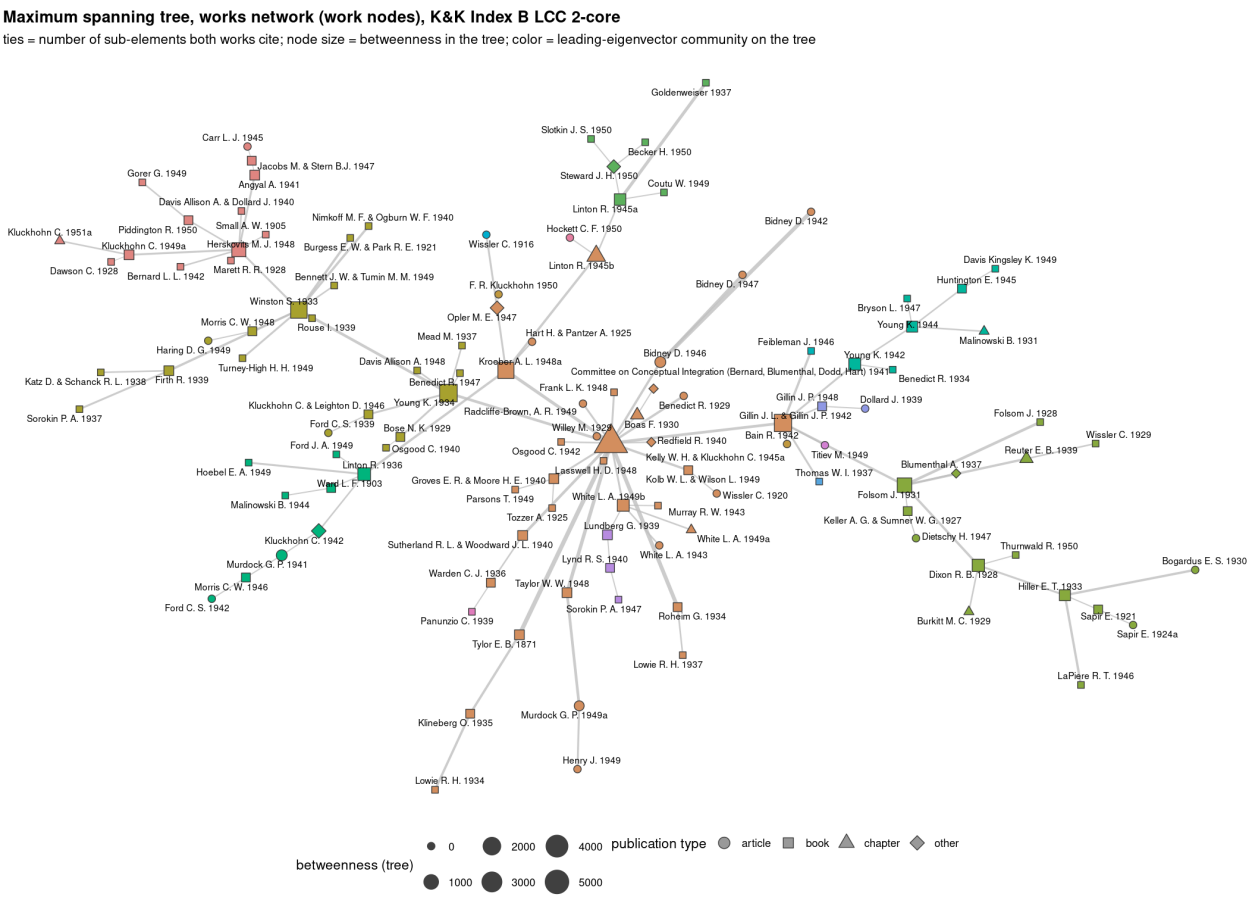

Maximum spanning tree, works network (work nodes), K&K Index B LCC 2-core
ties = number of sub-elements both works cite; node size = betweenness in the tree; color = leading-eigenvector community on the tree
Golderweiser 1937
Slotkin J. S. 1950
Becker H. 1950
Steward J. H. 1950
Coutu W. 1949
Linton R. 1945a
Hockett C. F. 1950
Linton R. 1945b
Bidney D. 1942
Bidney D. 1947
Carr L. J. 1945
Jacobs M. & Stern B.J. 1947
Angyal A. 1941
Gorer G. 1949
Davis Allison A. & Dollard J. 1940
Small A. W. 1905
Nimkoff M. F. & Ogburn W. F. 1940
Burgess E. W. & Park R. E. 1921
Wissler C. 1916
Kluckhohn C. 1951a
Piddington R. 1950
Kluckhohn C. 1949a
Herskovits M. J. 1948
Dawson C. 1928
Bernard L. L. 1942
Marett R. R. 1928
Bennett J. W. & Tumin M. M. 1949
F. R. Kluckhohn 1950
Winston S. 1933
Morris C. W. 1948
Rouse I. 1939
Opler M. E. 1947
Mead M. 1937
Hart H. & Pantzer A. 1925
Haring D. G. 1949
Turney-High H. H. 1949
Davis Allison A. 1948
Kroeber A. L. 1948a
Katz D. & Schanck R. L. 1938
Firth R. 1939
Sorokin P. A. 1937
Benedict R. 1947
Bidney D. 1946
Committee on Conceptual Integration (Bernard, Blumenthal, Dodd, Hart) 1941
Frank L. K. 1948
Kluckhohn C. & Leighton D. 1946
Young K. 1934
Radcliffe-Brown, A. R. 1949
Benedict R. 1929
Ford C. S. 1939
Bose N. K. 1929
Wiley M. 1929
Boas F. 1930
Ford J. A. 1949
Osgood C. 1940
Osgood C. 1942
Redfield R. 1940
Linton R. 1936
Hoebel E. A. 1949
Ward L. F. 1903
Groves E. R. & Moore H. E. 1940
Lasswell H. D. 1948
Kelly W. H. & Kluckhohn C. 1945a
Kolb W. L. & Wilson L. 1949
Malinowski B. 1944
Parsons T. 1949
White L. A. 1949b
Wissler C. 1920
Tozzer A. 1925
Murray R. W. 1943
Kluckhohn C. 1942
Lundberg G. 1939
White L. A. 1949a
Sutherland R. L. & Woodward J. L. 1940
White L. A. 1943
Murdock G. P. 1941
Lynd R. S. 1940
Warden C. J. 1936
Morris C. W. 1946
Taylor W. W. 1948
Ford C. S. 1942
Sorokin P. A. 1947
Roheim G. 1934
Panunzio C. 1939
Tylor E. B. 1871
Lowie R. H. 1937
Kluckhohn C. 1945a
Thomas W. I. 1937
Titiev M. 1949
Bain R. 1942
Gillin J. P. 1948
Gillin J. L. & Gillin J. P. 1942
Dollard J. 1939
Feibleman J. 1946
Young K. 1942
Benedict R. 1934
Davis Kingsley K. 1949
Huntington E. 1945
Bryson L. 1947
Young K. 1944
Malinowski B. 1931
Folsom J. 1928
Wissler C. 1929
Reuter E. B. 1939
Blumenthal A. 1937
Folsom J. 1931
Keller A. G. & Sumner W. G. 1927
Dietschy H. 1947
Thurnwald R. 1950
Bogardus E. S. 1930
Dixon R. B. 1928
Hiller E. T. 1933
Burkitt M. C. 1929
Sapir E. 1921
Sapir E. 1924a
LaPiere R. T. 1946
Kluckhohn C. 1951a
Kluckhohn C. 1949a
Kluckhohn C. & Leighton D. 1946
Kluckhohn C. 1942
Kluckhohn C. 1945a
Kluckhohn C. & Leighton D. 1946
Kluckhohn C. 1942
Kluckhohn C. 1945a
Kluckhohn C. 1949a
Kluckhohn C. 1951

Table 2: Top 10 nodes by Freeman-normalized betweenness in the definitions network MST.

Node type	Node	Norm. Betw. (%)	Degree
Concepts	social	69.11	16
	group	62.73	12
	society	49.06	14
	behavior	48.88	6
	customs	31.19	6
	total	18.76	3
	beliefs	17.89	6
	systems	14.51	6
	sum	12.01	3
	members of a society	10.88	3
Definitions	F-I-3	69.14	4
	C-I-6	57.54	12
	F-I-8	48.46	6
	F-II-8	40.84	4
	F-I-19	38.60	6
	A-3	34.04	9
	C-II-3	33.15	3
	C-II-4	29.35	4
	A-16	26.62	3
	C-I-4	22.19	3

Table 3: Top 10 nodes by Freeman-normalized betweenness in the works network MST.

Node type	Node	Norm. Betw. (%)	Degree
Concepts	social	73.18	15
	group	56.51	11
	behavior	50.04	6
	society	49.79	22
	beliefs	33.57	16
	customs	33.33	4
	ideas	19.09	4
	total	14.71	7
	generations	11.07	4
	members of a society	9.79	2
Works	Kelly & Kluckhohn 1945a	83.02	19
	Young 1934	41.78	7
	Gillin 1942	40.14	8
	Winston 1933	34.13	9
	Kroeber 1948a	31.58	5
	Folsom 1931	21.98	6
	Herskovits 1948	19.28	8
	Linton 1936	13.05	5
	Dixon 1928	11.42	4
	Linton 1945b	11.34	3